

CHAPTER 2:

Atomic Structure and Interatomic Bonding

ISSUES TO ADDRESS...

- What promotes bonding?
- What types of bonds are there?
- What properties are inferred from bonding?



Atomic Structure (Freshman Chem.)

- atom – electrons – 9.11×10^{-31} kg
 protons
 neutrons
- atomic number = # of protons in nucleus of atom
 = # of electrons of neutral species
- A [=] atomic mass unit = amu = 1/12 mass of ^{12}C

Atomic wt = wt of 6.022×10^{23} molecules or atoms

$$1 \text{ amu/atom} = 1 \text{ g/mol}$$

C 12.011
H 1.008 etc.

AVAGADRO'S NUMBER = $6.022 \times 10^{23} = N_A$

**ATOMIC OR MOLECULAR WEIGHT =
 $N_A \times$ WEIGHT PER ATOM.**

number of neutrons = N

number of protons = Z

$A = Z + N$

(2.1)

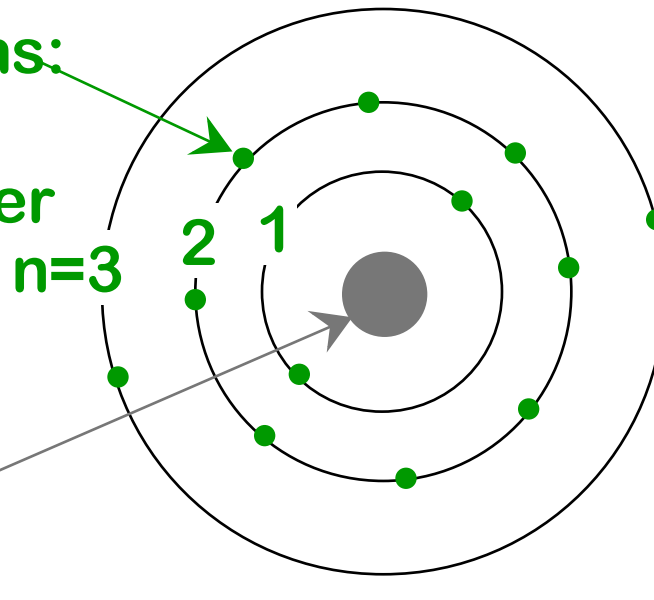


Atomic Structure

- Valence electrons determine all of the following properties
 - 1) Chemical
 - 2) Electrical
 - 3) Thermal
 - 4) Optical

BOHR ATOM

orbital electrons:
 n = principal
quantum number



Adapted from Fig. 2.1,
Callister 6e.

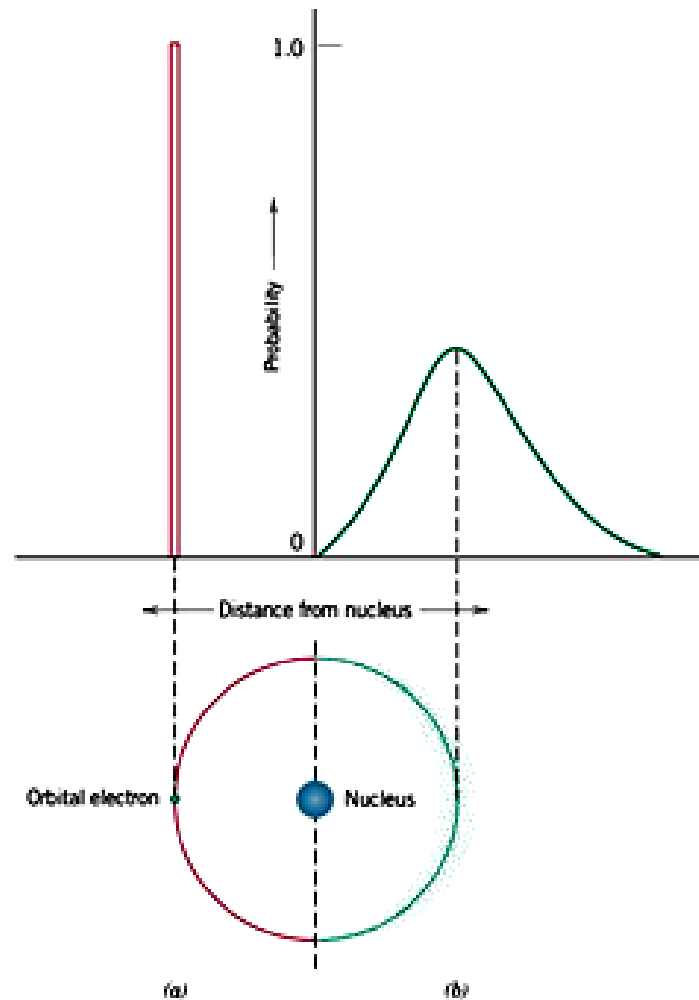
Nucleus: Z = # protons

= 1 for hydrogen to 94 for plutonium

N = # neutrons

Atomic mass $A \approx Z + N$

WAVE MECHANICAL MODEL OF ATOM



Electronic Structure

- Electrons have wavelike and particulate properties.
 - This means that electrons are in **orbitals** defined by a probability.
 - Each orbital at discrete energy level is determined by **quantum numbers**.

Quantum

n = principal (energy level-shell)

l = subsidiary (orbitals)

m_l = magnetic

m_s = spin

Designation

K, L, M, N, O (1, 2, 3, etc.)

s, p, d, f (0, 1, 2, 3, ..., $n-1$)

1, 3, 5, 7 ($-l$ to $+l$)

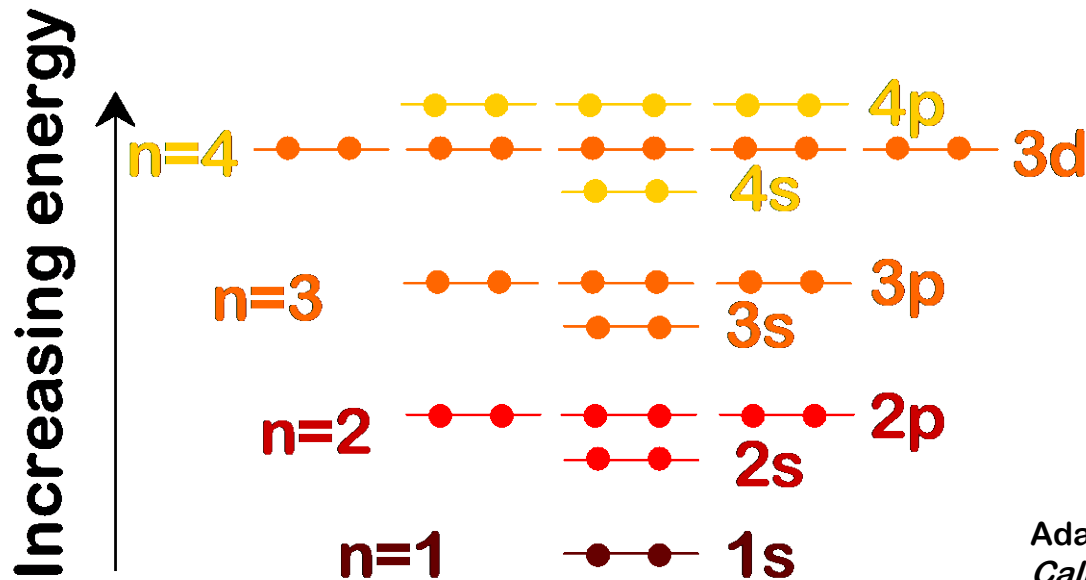
$\frac{1}{2}, -\frac{1}{2}$



ELECTRON ENERGY STATES

Electrons...

- have discrete **energy states**
- tend to occupy lowest available energy state.



Adapted from Fig. 2.5,
Callister 6e.

STABLE ELECTRON CONFIGURATIONS

Stable electron configurations...

- have complete s and p subshells
- tend to be **unreactive**.

Z	Element	Configuration
---	---------	---------------

2	He	$1s^2$
---	----	--------

10	Ne	$1s^2 2s^2 2p^6$
----	----	------------------

18	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$
----	----	----------------------------

36	Kr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$
----	----	--

Adapted from Table 2.2,
Callister 6e.



SURVEY OF ELEMENTS

- Most elements: Electron configuration **not stable**.

Element	Atomic #	Electron configuration
---------	----------	------------------------

Hydrogen	1	$1s^1$
----------	---	--------

Helium	2	$1s^2$ (stable)
--------	---	-----------------

Lithium	3	$1s^2 2s^1$
---------	---	-------------

Beryllium	4	$1s^2 2s^2$
-----------	---	-------------

Boron	5	$1s^2 2s^2 2p^1$
-------	---	------------------

Carbon	6	$1s^2 2s^2 2p^2$
--------	---	------------------

Adapted from Table 2.2,
Callister 6e.

...

...

Neon	10	$1s^2 2s^2 2p^6$ (stable)
------	----	---------------------------

Sodium	11	$1s^2 2s^2 2p^6 3s^1$
--------	----	-----------------------

Magnesium	12	$1s^2 2s^2 2p^6 3s^2$
-----------	----	-----------------------

Aluminum	13	$1s^2 2s^2 2p^6 3s^2 3p^1$
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...

...

Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$ (stable)
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...

...

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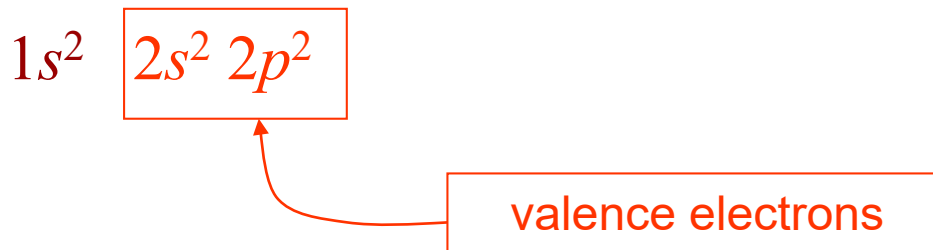
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$ (stable)
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- Why? **Valence** (outer) shell usually not filled completely.



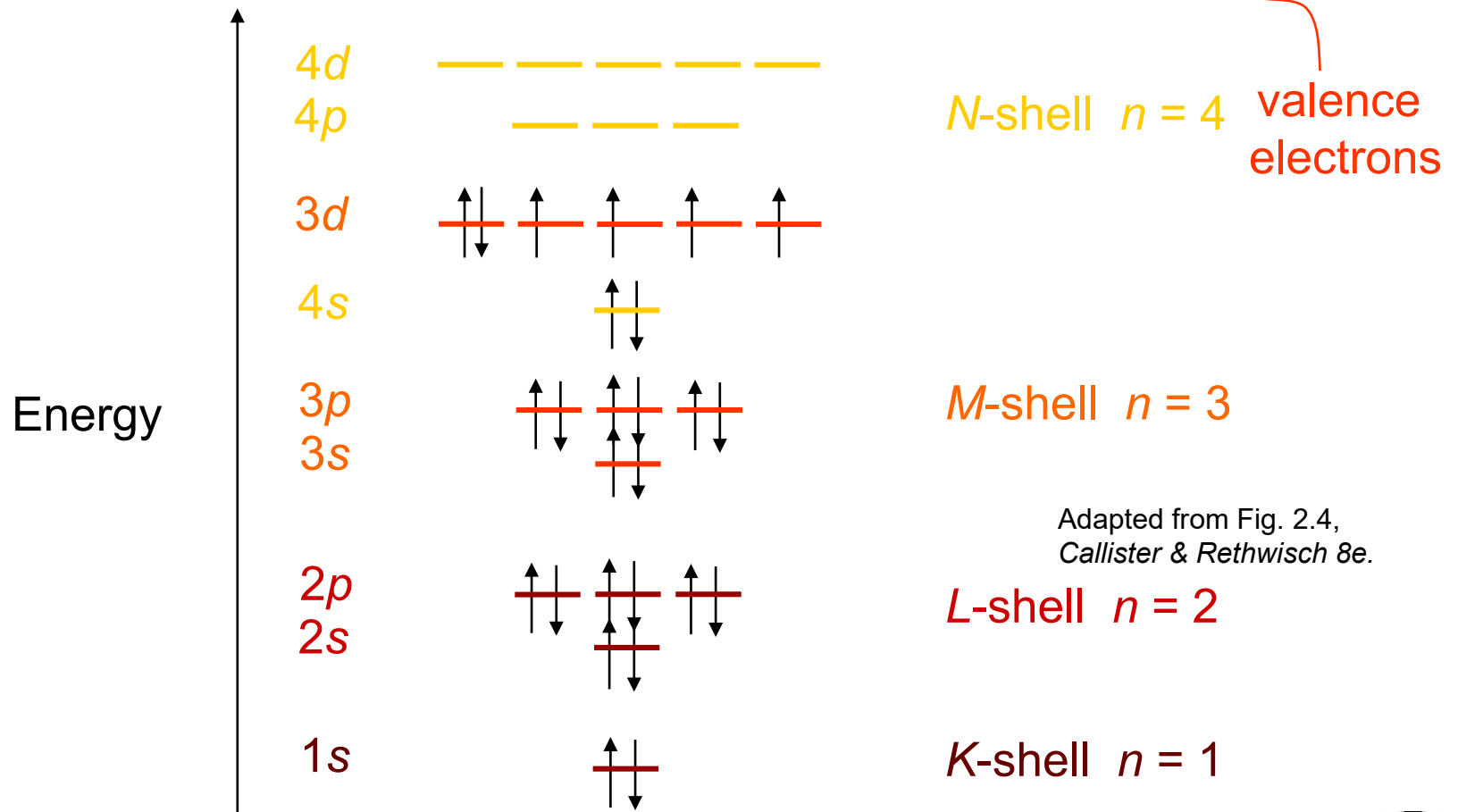
Electron Configurations

- Valence electrons – those in unfilled shells
 - Filled shells more stable
 - Valence electrons are most available for bonding and tend to control the chemical properties
- example: C (atomic number = 6)



Electronic Configurations

ex: Fe - atomic # = 26 $1s^2 2s^2 2p^6 3s^2 3p^6$ $3d^6 4s^2$



Adapted from Fig. 2.4,
Callister & Rethwisch 8e.



THE PERIODIC TABLE

- Columns: Similar **Valence** Structure

Key

- Atomic number
- Symbol
- Atomic weight

Metal
 Nonmetal
 Intermediate

1 H 1.01	2 He 4.00																
3 Li 6.94	4 Be 9.01																
11 Na 22.99	12 Mg 24.31																
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 51.99	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.91	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57-71 Rare earth series	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.19	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89-103 Actinide series															

give up 1e
 give up 2e
 give up 3e

accept 2e
 accept 1e
 inert gases

Adapted from Fig. 2.6, Callister 6e.

Electropositive elements:
Readily give up electrons
to become + ions.

Electronegative elements:
Readily acquire electrons
to become - ions.



ELECTRONEGATIVITY

- Ranges from 0.7 to 4.0,
- Large values: tendency to acquire electrons.

IA H 2.1	IIA											IIIA	IVA	VA	VIA	VIIA	0 He -
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne -
Na 0.9	Mg 1.2	IIIB	IVB	VB	VIB	VIIIB	VIII			IB	IIIB	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar -
K 0.8	Ca 1.0	21 Sc 1.3	22 Ti 1.5	23 V 1.6	24 Cr 1.6	25 Mn 1.5	26 Fe 1.8	27 Co 1.8	28 Ni 1.8	29 Cu 1.9	30 Zn 1.8	31 Ga 1.6	32 Ge 1.8	33 As 2.0	34 Se 2.4	35 Br 2.8	36 Kr -
Rb 0.8	Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5	54 Xe -
Cs 0.7	Ba 0.9	57-71 La-Lu 1.1-1.3	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	85 At 2.2	86 Rn -
Fr 0.7	Ra 0.9	89-103 Ac-No 1.1-1.7															



Smaller electronegativity

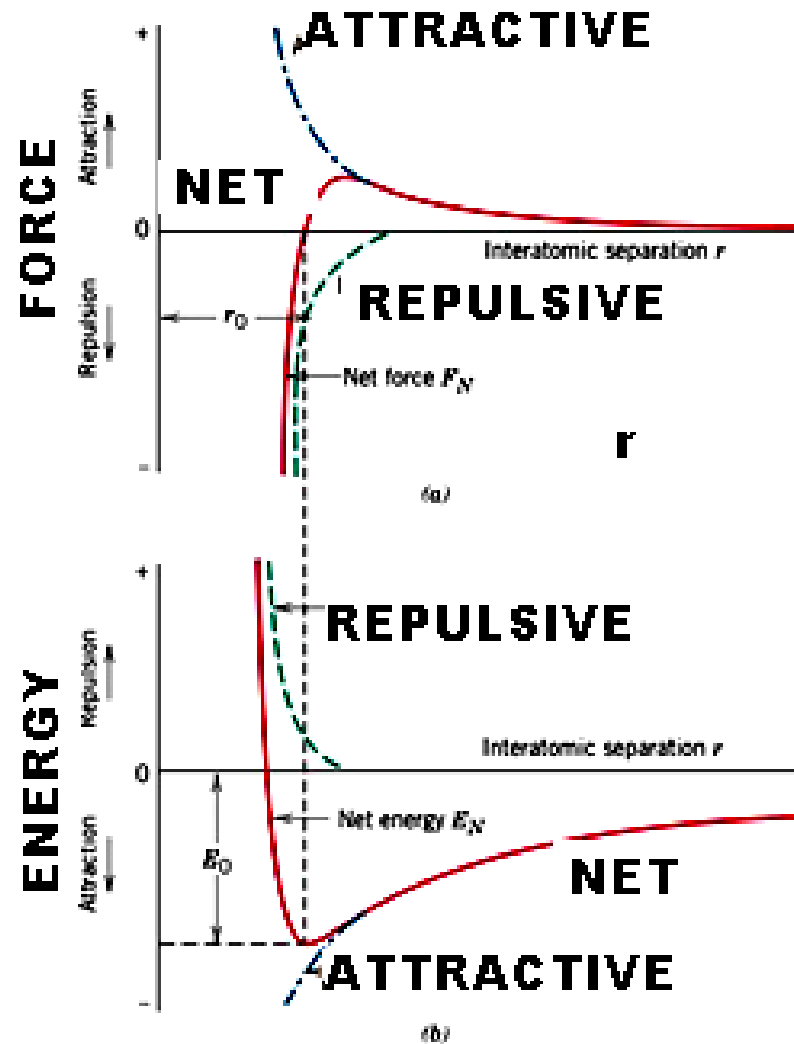


Larger electronegativity

Adapted from Fig. 2.7, *Callister 6e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

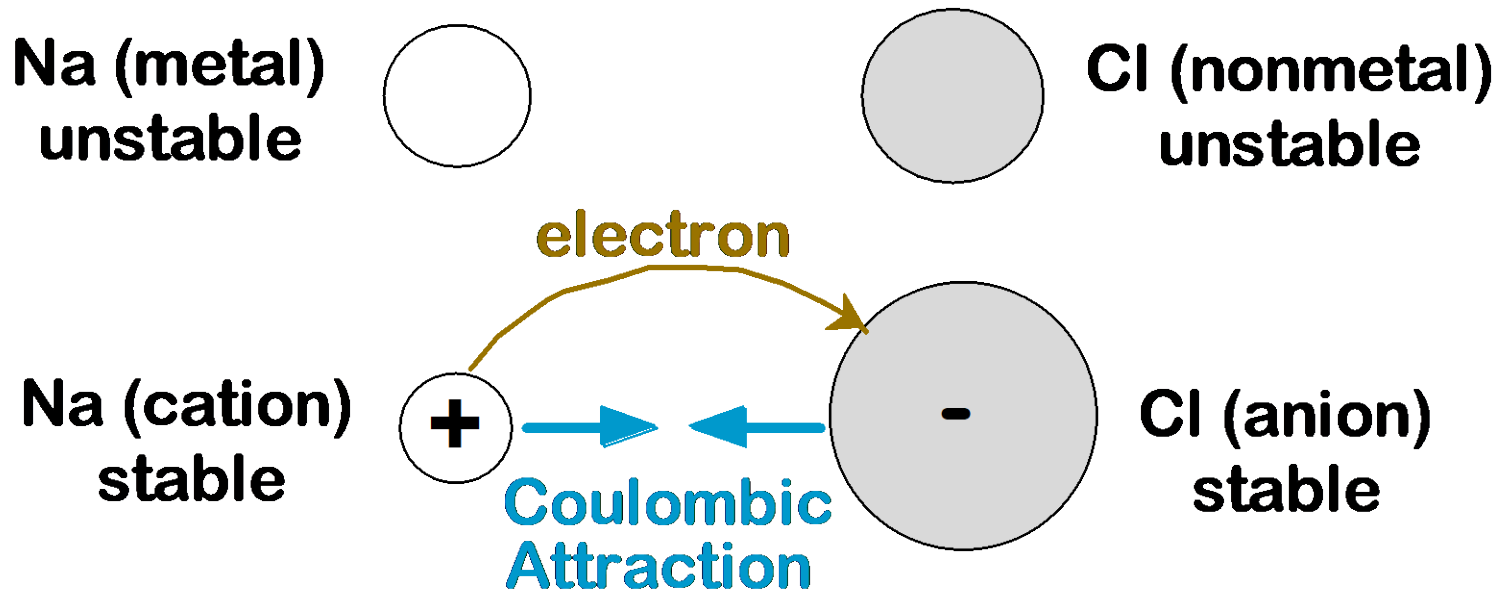


FORCES AND ENERGIES



IONIC BONDING

- Occurs between + and - ions.
- Requires **electron transfer**.
- Large difference in electronegativity required.
- Example: NaCl



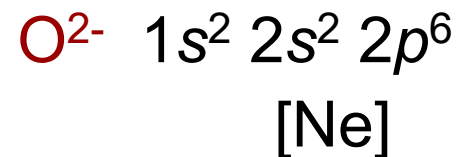
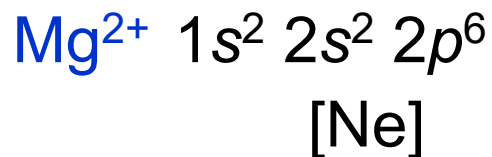
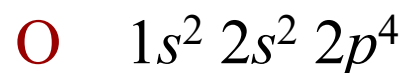
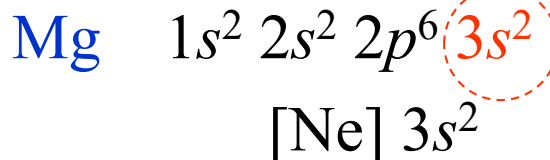
Ionic bond – metal + nonmetal

↑
donates
electrons

↑
accepts
electrons

Dissimilar electronegativities

ex: MgO



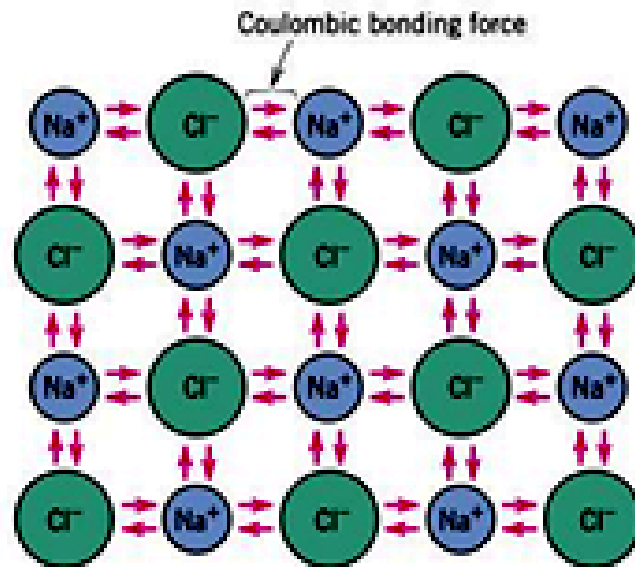
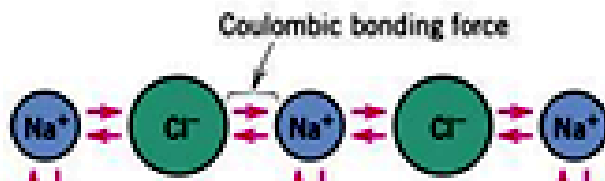


Figure 2.9
bonding in :



FORCES, ATTRACTION F_A

Bonding Forces and Energies

2.13 Calculate the force of attraction between a K^+ and an O^{2-} ion the centers of which are separated by a distance of $r_0 = 1.5 \text{ nm}$.

Solution

The attractive force between two ions F_A is just the derivative with respect to the interatomic separation of the attractive energy expression, Equation 2.8, which is just



$$F_A = \frac{dE_A}{dr} = \frac{d\left(-\frac{A}{r}\right)}{dr} = \frac{A}{r^2}$$

The constant A in this expression is defined in footnote 3. Since the valences of the K^+ and O^{2-} ions

(Z_1 and Z_2) are +1 and -2, respectively, $Z_1 = 1$ and $Z_2 = 2$, then

$$F_A = \frac{(Z_1 e)(Z_2 e)}{4\pi\epsilon_0 r^2}$$

$$= \frac{(1)(2)(1.602 \times 10^{-19} \text{ C})^2}{(4)(\pi)(8.85 \times 10^{-12} \text{ F/m})(1.5 \times 10^{-9} \text{ m})^2}$$

$$= 2.05 \times 10^{-10} \text{ N}$$

IONIC FORCE / P 31 FOOT-NOTE

$$F = (Z_1 * Z_2 * e^2) / (4 * \pi * \epsilon_0 * r^2);$$

$$e = 1.602 * 10^{(-19)} \text{ COULOMBS ;}$$

$$\epsilon_0 = 8.85 * 10^{(-12)}$$

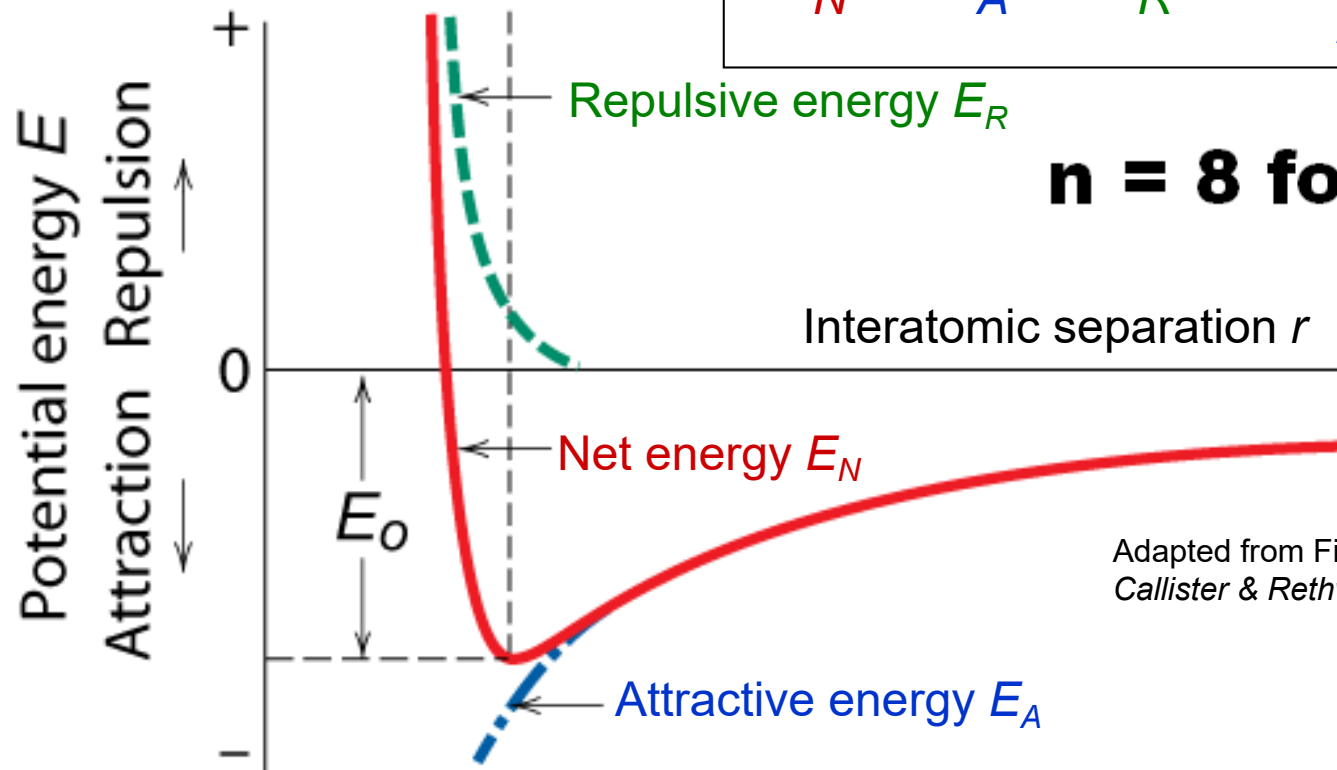
Z_1, Z_2 = VALENCIES OF IONS

Ionic Bonding

- Energy – minimum energy most stable
 - Energy balance of attractive and repulsive terms

$$E_N = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}$$

$n = 8$ for ions



Adapted from Fig. 2.8(b),
Callister & Rethwisch 8e.

EXAMPLES: IONIC BONDING

- Predominant bonding in **Ceramics**

NaCl

MgO

CaF₂

CsCl

IA																	0						
H 2.1																	He -						
Li 1.0	IIA	Be 1.5															Ne -						
Na 0.9	Mg 1.2															Ar -							
																		III A	IV A	V A		VI A	
																		B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	
																		Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.8	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr -						
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe -						
Cs 0.7	Ba 0.9	La-Lu 1.1-1.2	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn -						
Fr 0.7	Ra 0.9	Ac-No 1.1-1.7																					

Give up electrons

Acquire electrons

Adapted from Fig. 2.7, *Callister 6e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.



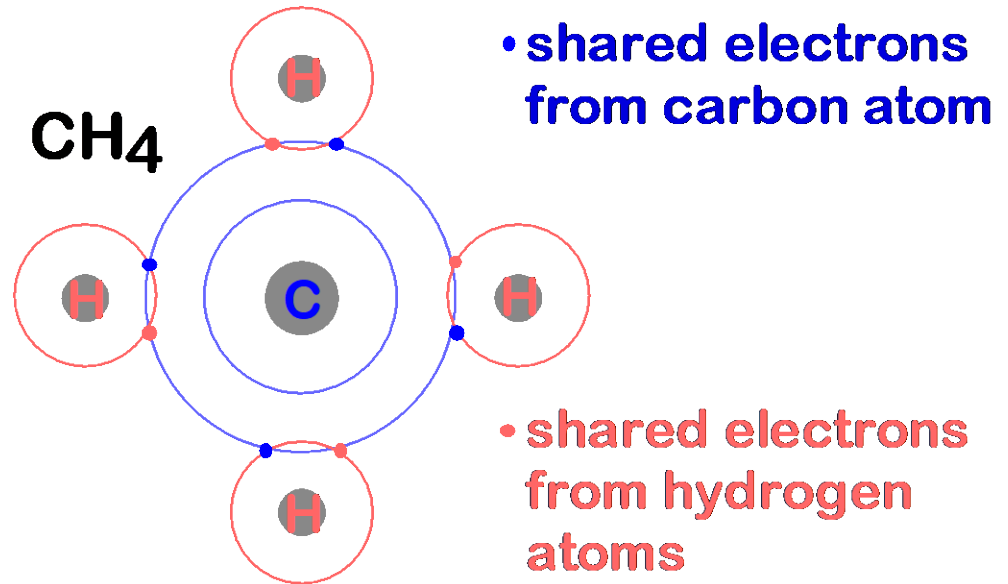
COVALENT BONDING

- Requires **shared electrons**
- Example: CH₄

C: has 4 valence e,
needs 4 more

H: has 1 valence e,
needs 1 more

Electronegativities
are comparable.



Adapted from Fig. 2.10, *Callister 6e*.

Primary Bonding

- Metallic Bond -- delocalized as electron cloud
- Ionic-Covalent Mixed Bonding

$$\% \text{ ionic character} = \left(1 - e^{-\frac{(X_A - X_B)^2}{4}} \right) \times (100\%)$$

where X_A & X_B are Pauling electronegativities

Ex: MgO

$$\begin{aligned} X_{\text{Mg}} &= 1.2 \\ X_{\text{O}} &= 3.5 \end{aligned}$$

$$\% \text{ ionic character} = \left(1 - e^{-\frac{(3.5 - 1.2)^2}{4}} \right) \times (100\%) = 73.4\% \text{ ionic}$$



EXAMPLES: COVALENT BONDING

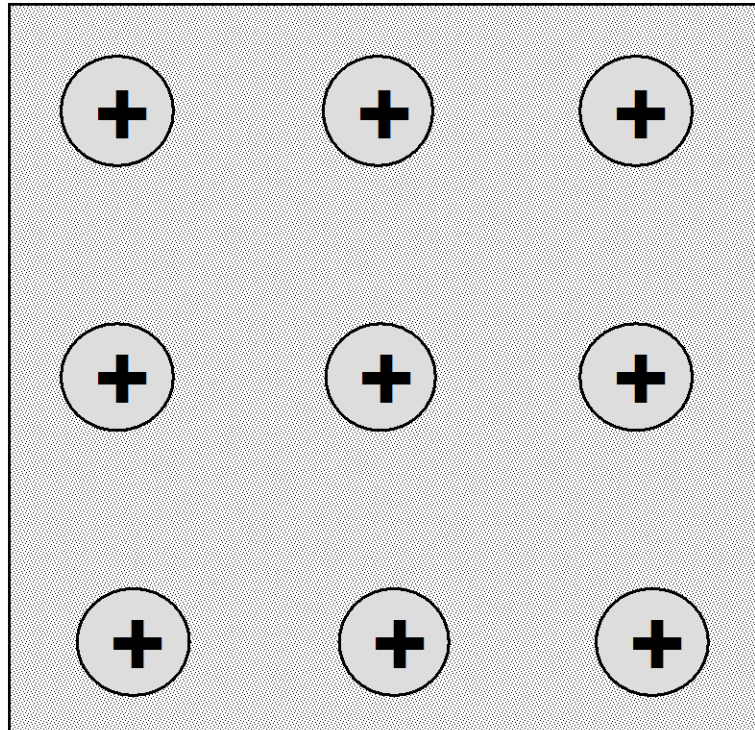
Adapted from Fig. 2.7, *Callister 6e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright

Adapted from Fig. 2.7, *Callister 6e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

- Molecules with **nonmetals**
- Molecules with **metals** and **nonmetals**
- Elemental solids (RHS of Periodic Table)
- Compound solids (about **column IVA**)

METALLIC BONDING

- Arises from a sea of **donated valence electrons** (1, 2, or 3 from each atom).



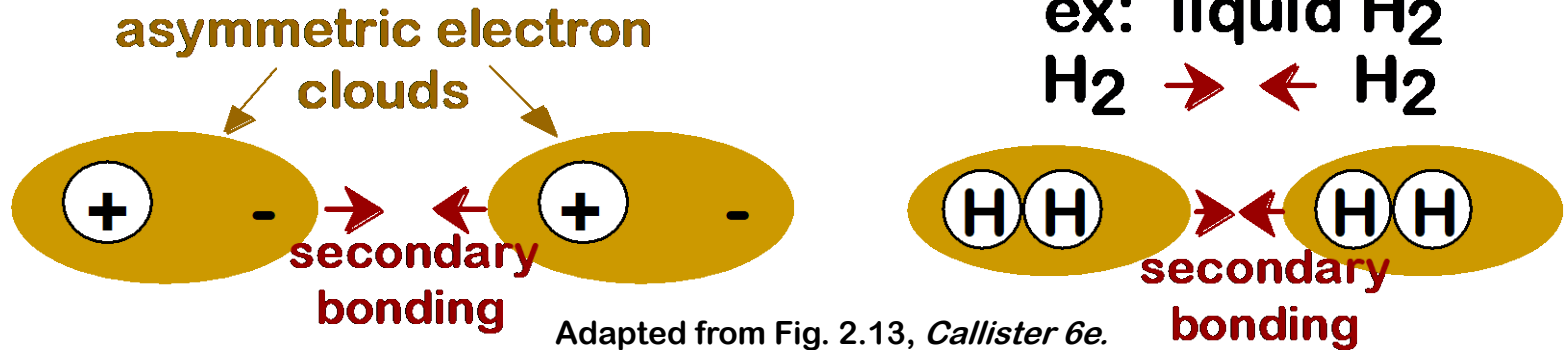
Adapted from Fig. 2.11, *Callister 6e*.

- Primary bond for **metals** and their **alloys**

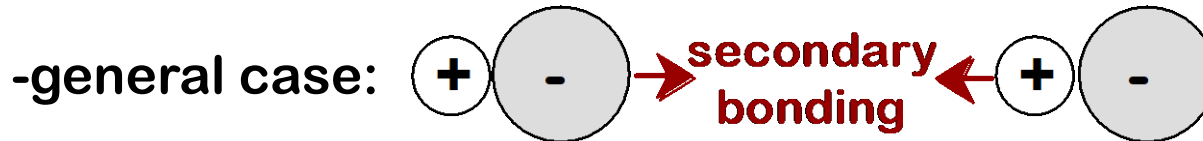
SECONDARY BONDING

Arises from interaction between **dipoles**

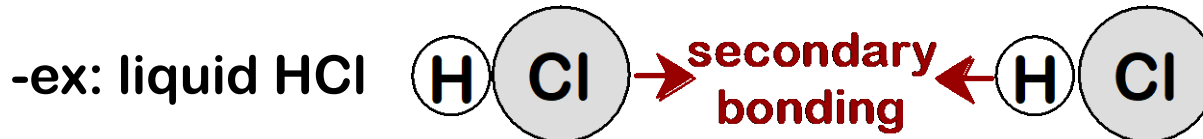
- Fluctuating **dipoles**



- Permanent **dipoles**-molecule induced

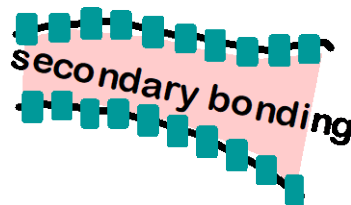


Adapted from Fig. 2.14, Callister 6e.



Adapted from Fig. 2.14, Callister 6e.

-ex: polymer



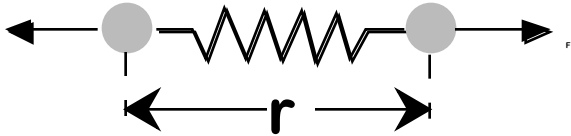
SUMMARY: BONDING

Type	Bond Energy	Comments
Ionic	Large!	Nondirectional (ceramics)
Covalent	Variable large-Diamond small-Bismuth	Directional semiconductors , ceramics polymer chains)
Metallic	Variable large-Tungsten small-Mercury	Nondirectional (metals)
Secondary	smallest	Directional inter-chain (polymer) inter-molecular

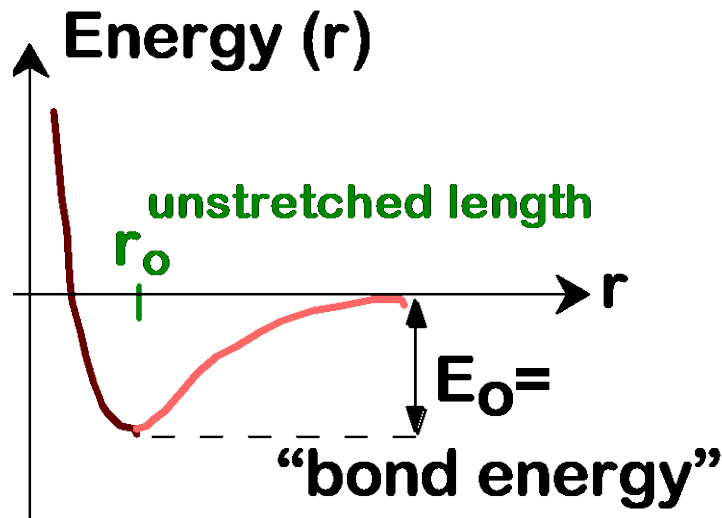


PROPERTIES FROM BONDING: T_M

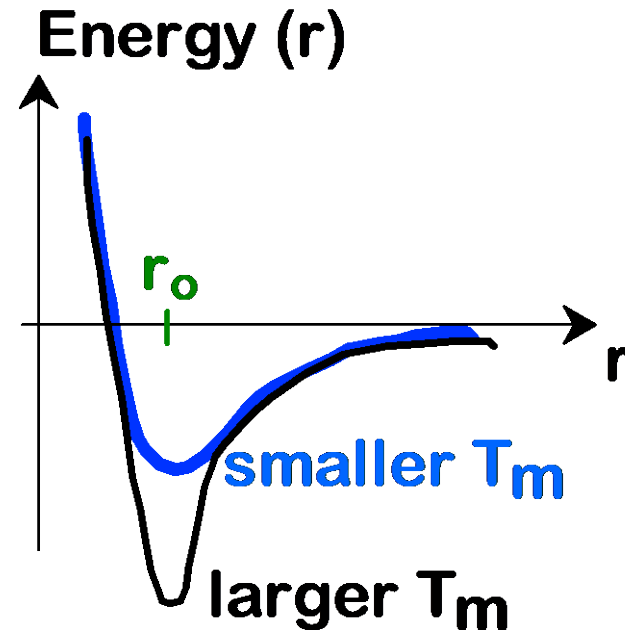
- Bond length, r



- Bond energy, E_o



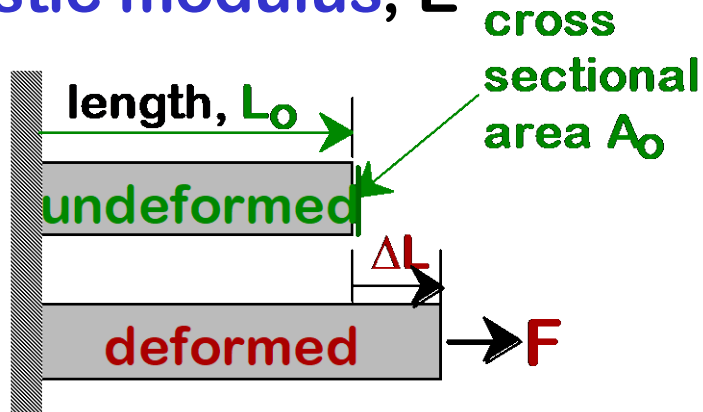
- Melting Temperature, T_m



T_m is larger if E_o is larger.

PROPERTIES FROM BONDING: E

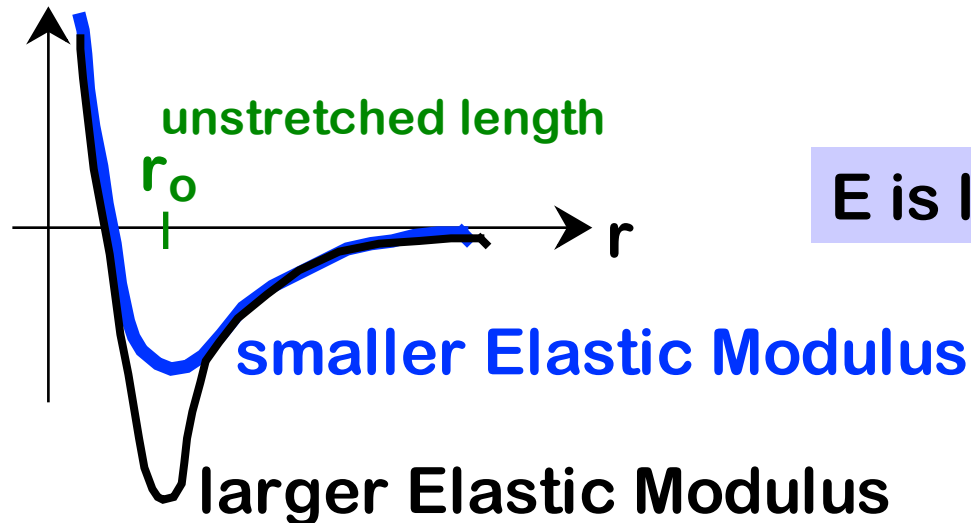
- Elastic modulus, E



Elastic modulus

$$\frac{F}{A_0} = E \frac{\Delta L}{L_0}$$

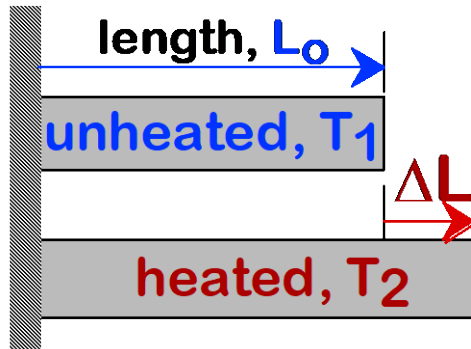
- $E \sim$ curvature at r_0
Energy



E is larger if E_0 is larger.

PROPERTIES FROM BONDING: α

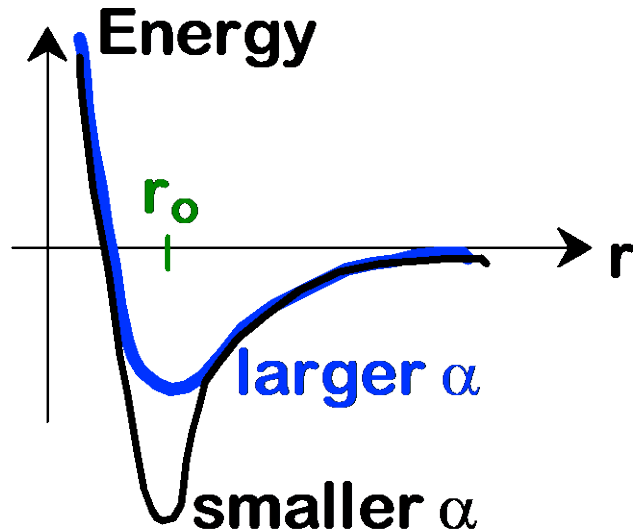
- Coefficient of thermal expansion, α



coeff. thermal expansion

$$\frac{\Delta L}{L_0} = \alpha (T_2 - T_1)$$

- $\alpha \sim$ symmetry at r_0



α is larger if E_0 is smaller.

SUMMARY: PRIMARY BONDS

Ceramics

(Ionic & covalent bonding):

Large bond energy

large T_m

large E

small α

Metals

(Metallic bonding):

Variable bond energy

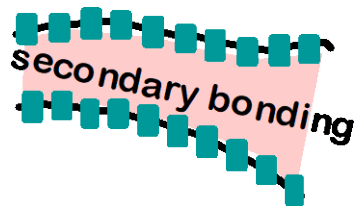
moderate T_m

moderate E

moderate α

Polymers

(Covalent & Secondary):



Directional Properties

Secondary bonding dominates

small T

small E

large α